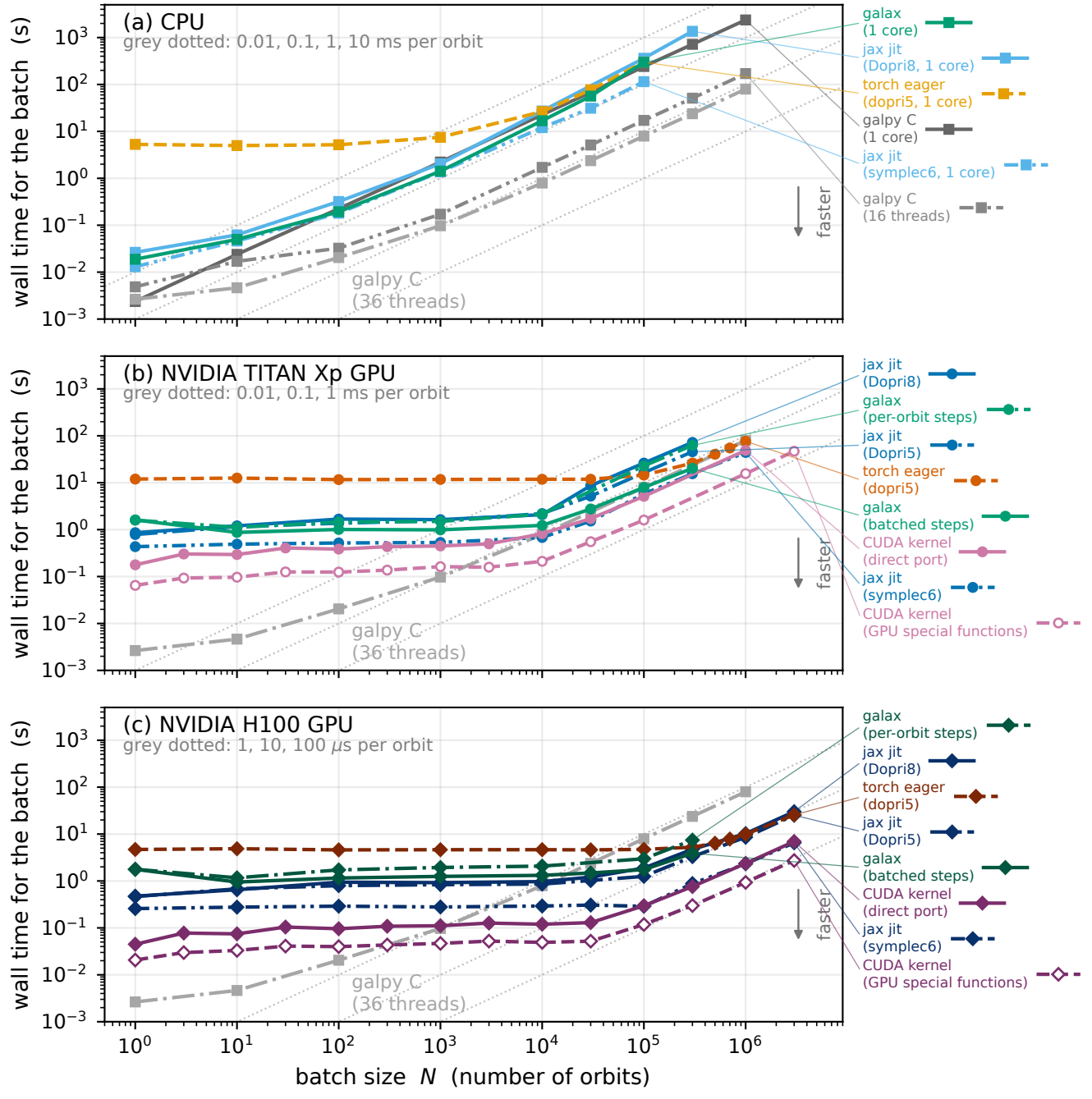
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per orbit
wall time for the batch (s)
galax (1 core)
jax jit (Dopri8, 1 core)
torch eager (dopri5, 1 core)
galpy C (1 core)
jax jit (symplec6, 1 core)
galpy C (16 threads)
faster
galpy C (36 threads)
(b) NVIDIA TITAN Xp GPU
grey dotted: 0.01, 0.1, 1 ms per orbit
wall time for the batch (s)
jax jit (Dopri8)
galax (per-orbit steps)
jax jit (Dopri5)
torch eager (dopri5)
galax (batched steps)
CUDA kernel (direct port)
jax jit (symplec6)
CUDA kernel (GPU special functions)
faster
galpy C (36 threads)
(c) NVIDIA H100 GPU
grey dotted: 1, 10, 100 μs per orbit
wall time for the batch (s)
galax (per-orbit steps)
jax jit (Dopri8)
torch eager (dopri5)
jax jit (Dopri5)
galax (batched steps)
CUDA kernel (direct port)
jax jit (symplec6)
CUDA kernel (GPU special functions)
faster
galpy C (36 threads)
batch size N (number of orbits)